MSE 204: Transport Phenomena in Materials Engineering

Course project on

Efficient oil- water separation using wax-coated sponge material

Submitted by

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Abstract

In the present study, we investigated the oil water separation ability of wax-coated sponge extracted from the packaging material. To characterize the sponge, we performed SEM and porosity measurements. Further, the wetting properties of groundnut oil, diesel and DI water were investigated for both coated and uncoated specimen. The water droplets roll-off at ~24° on the wax coated sponge, whereas oil and diesel droplets get absorbed. In addition, the oil collection at different oil-water concentration is investigated. We found that the coated sponge has high oil absorption capacity of 28.83 g/g and 23.15 g/g for groundnut oil and diesel. In addition, the developed superhydrophobic sponge can easily remove the diesel from 3.5 wt. % saltwater.

Introduction and Literature survey

Oil spill in the sea is a major environmental concern, as they cause significant harm to marine life. Therefore, effective methods for separating crude oil from seawater are essential to mitigate the negative impacts on environment. Traditional methods lack efficiency and safety, thus current research efforts are going on to develop better methods. In recent years, superhydrophobic sponges that are water-repellent and oleophilic have gained attention as promising solution for oil-water separation. They have high porosity, large surface area and good affinity with water due to which they can float and simultaneously absorb oil and therefore useful in treating oil spills in the ocean. They also exhibit high absorption capacity and reusability which makes it a cost-effective solution.

We have read many literatures which talks about oil-water separation. Some studies focused on collecting water from oil, while others focused on collecting oil from water. Some of the literature on oil water separation are tabulate in the following Table. Different oils such as silicone oil, motor oil, hexadecane etc are used for the study. All the modified sponge showed high oil absorption capacity as shown the table below.

Table: Literature on different superhydrophobic sponges.

S.no	Sponge Material	Coating Material	Porosity	Oil used	Oil Absorption capacity	Reference

1	Natural Sponge (Local Market)	Rice bran Wax and Candelilla wax	90%	Hexa Decane	16.7g/gm	[1]
2	PU Sponge	Steric acid	90.5%	N-hexane, cholorofor m paraffin oil	17.4 to 41.6 g/g	[2]
3	Melamine sponge blocks	carnauba wax and beeswax	99%		95ml/gm	[3]
4	Melamine sponge	Fe3O4/CS-PFNA solution	99%	Soyabean oil, silicone oil	Separtion efficiency= 96%	[4]
5	Polyuretha ne sponges (Density= 0.032 g/cm3)	1H,1H,2H,2H-Perfluorooctyltriethoxysilane (PTES), 0.3 M ferric chloride and 0.02 M PTES	97%	Motor oil, pump oil, silicone oil	20 times its own weigh	[5]

1. Experimental details

1.1. Materials

Sponge is sourced from packaging materials used for packing expensive instruments. Wax is extracted from candles purchased from the market. Groundnut oil and diesel is also obtained from the local market. Deionized (DI) water is collected from the laboratory.

1.2. Wax coated sponge preparation

The wax coated sponge is prepared using dip coating method. First the wax-hexane solution is prepared by adding 1.8 gm of candle wax in 300 ml of hexane which is heated at 60 °C for 60 min. Further the sponge (3 cm × 3 cm × 3 cm) is immersed in the prepared solution. In each cycle, sponge was immersed using stirring rod, then squeezed in the emulsion for 2 sec. Then

the sample is taken out and cooled and dried at the room temp for 5 min. Similarly, for each specimen, 4 cycle were repeated to obtain 8 superhydrophobic sponge as shown in Figure 1.

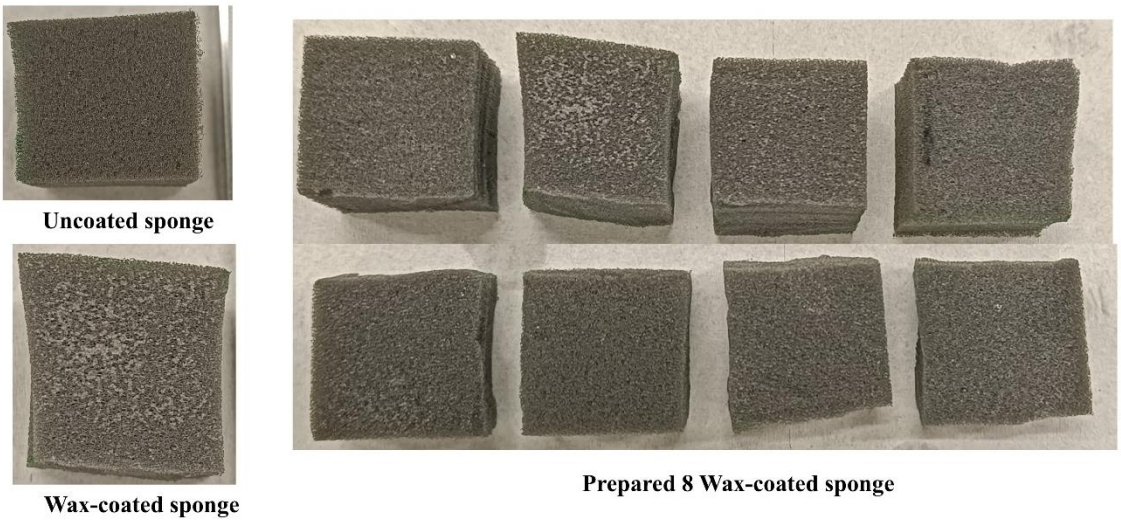


Figure 1. Digital images of uncoated and coated sponge.

1.3. Scanning electron microscopy (SEM) investigations

The surface features were analysed using a JEOL JSM7600F scanning electron microscope (SEM) operating at an accelerating voltage of 5 keV. A thin gold coating was applied to the sponge for 90 seconds. Images were then captured at various locations in secondary electron mode.

1.4. Wetting measurements

The static contact angles (SCAs) and roll-off angles (RAs) were measured using the sessile drop method with a contact angle goniometer (DSA25, KRÜSS, Germany). Deionized (DI) water droplets (5 μL) and groundnut oil droplets (5 μL) were used for these measurements.

1.5. Porosity measurements and Oil water separation

The open porosity of the sponge was measured using the gravimetric method. In this, first the dry weight of the sponge is measured. Then, sponge was immersed in DI water for 12 hours for complete saturation. Then wet weight is measured. Then the sponge is immersed in the DI water as shown in Figure 2, and immersed weight is measured.

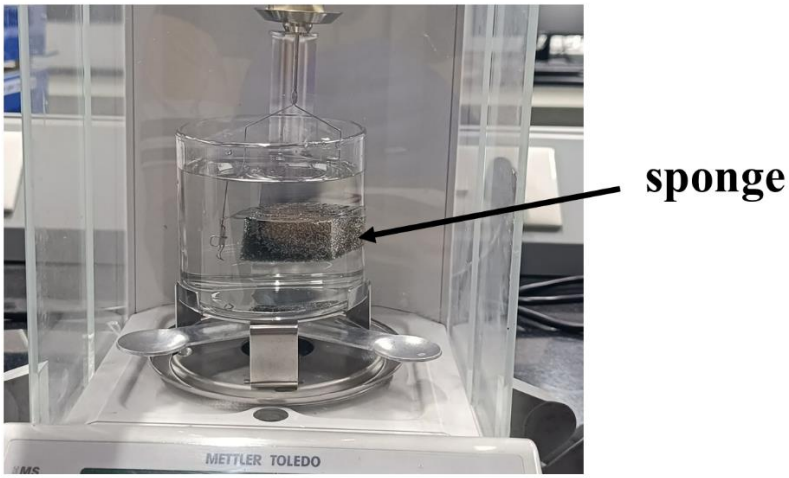


Figure 2. Density kit setup for porosity measurement.

2. Result and discussion

2.1. Surface microstructure of pristine sponge and wax-coated sponge

The surface structure of both coated and uncoated sponge were imaged as shown in Figure 3. The pores are elliptical in shape with average length and breadth of 334±88 µm and 206±31 µm which are measured using ImageJ. Considering the pores to be in spherical shape the pore radius is 270±90 µm, which will be used in further analysis. As shown in figure 3, the coated sponge consists of candle wax deposited over it.

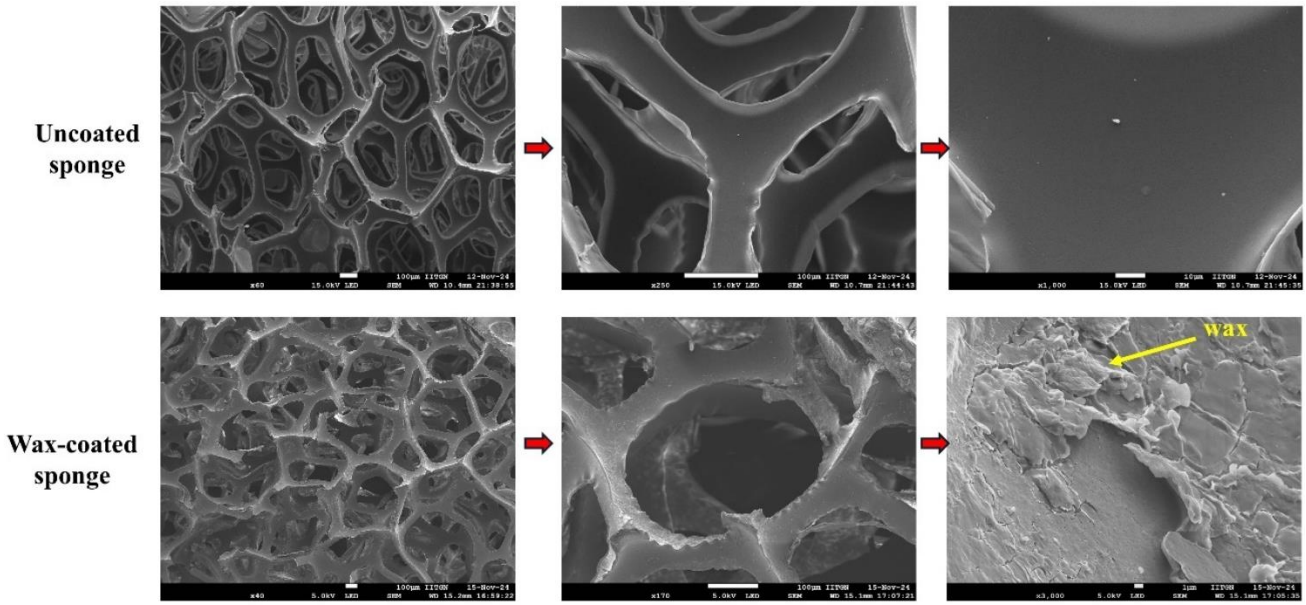


Figure 3. SEM images of coated and uncoated sponge.

2.2. Porosity measurements

The porosity of the sponge is measured using the equation below [6]. Two samples were taken for porosity measurements. Different measured weights for sponge and calculated porosity is shown in Table1. The average porosity is ~ 90%. The immersed weight is measured to be negative; this is due to the sponge sample. We are using sponge which has very low density due to which the buoyancy forces are higher than weight of the sponge resulting in a negative immersed weight. The porosity can be measured using the following equation,

$$\text{Porosity} = \frac{Wet\ weight - Dry\ weight}{Wet\ weight - Immersed\ weight}$$

Table 1. Porosity calculations

Sample	**Dry weight (gm)**	**Wet weight (gm)**	**Immersed weight (gm)**	**Porosity (%)**
1	0.49747	16.7758	-0.8223	92.4
2	0.12135	3.66786	-0.37006	87.81
Average porosity				90.105±3

2.3.Wetting measurements

Figure 4 shows the digital images and goniometer images of oil and water droplets on coated and uncoated sponge. The superhydrophobic sponge float on the water while uncoated submerged. Further, diesel, water and oil get absorbed in uncoated sponge. On coated sponge, only diesel, and oil get absorbed, while water droplets roll-off. The SCAs of oil and water droplets are shown in Figure 4 on both coated and uncoated sponge. The water droplets roll-off with at ~24° on coated sponge.

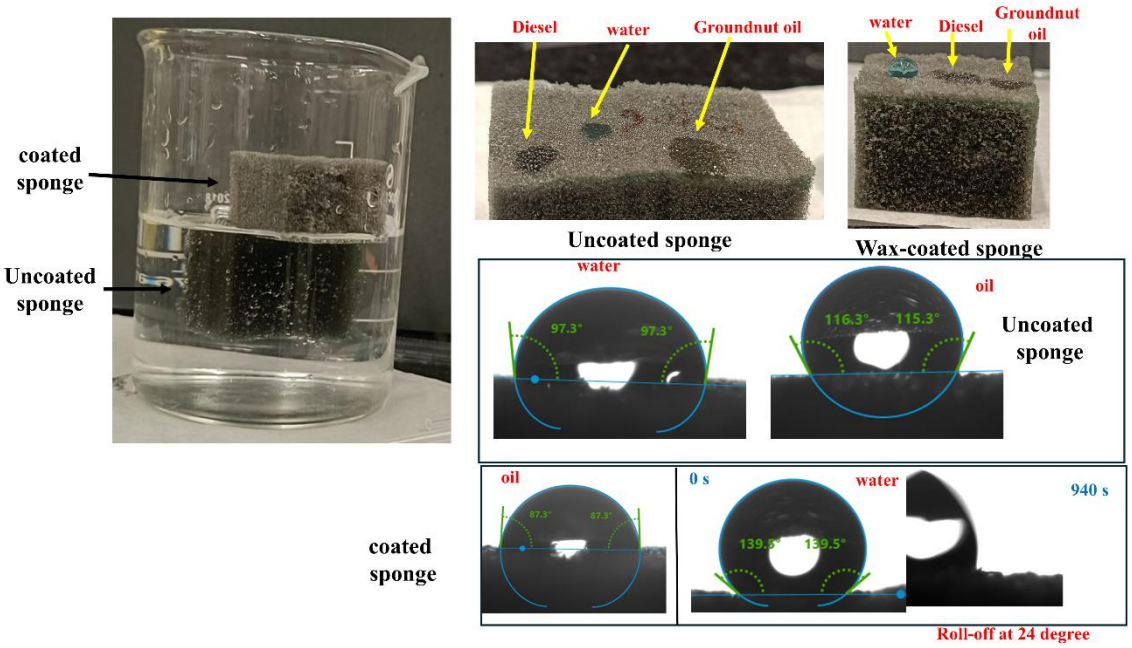
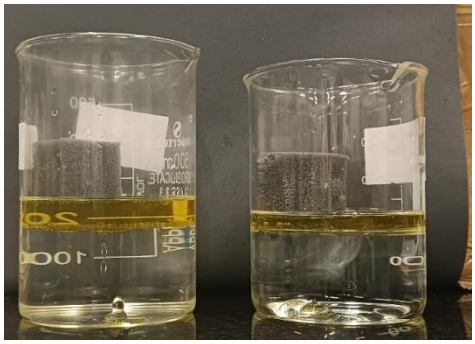


Figure 4. Digital images and goniometer images of oil and water droplets on coated and uncoated sponge.

2.4. Oil collection from different oil-water emulsions

For this study, we took two concentrations of oil in water i.e. 100 ml oil in 200 ml DI water and 40 ml oil in 200 ml water. Figure 5 shows the oil collection as a function of time for both concentrations. In lower concentration, ~17 gm of oil gets absorbed in ~320 min, while in higher concentration 20 gm of oil get absorbed in 225 min, which implies that oil water absorption is higher in higher concentration.



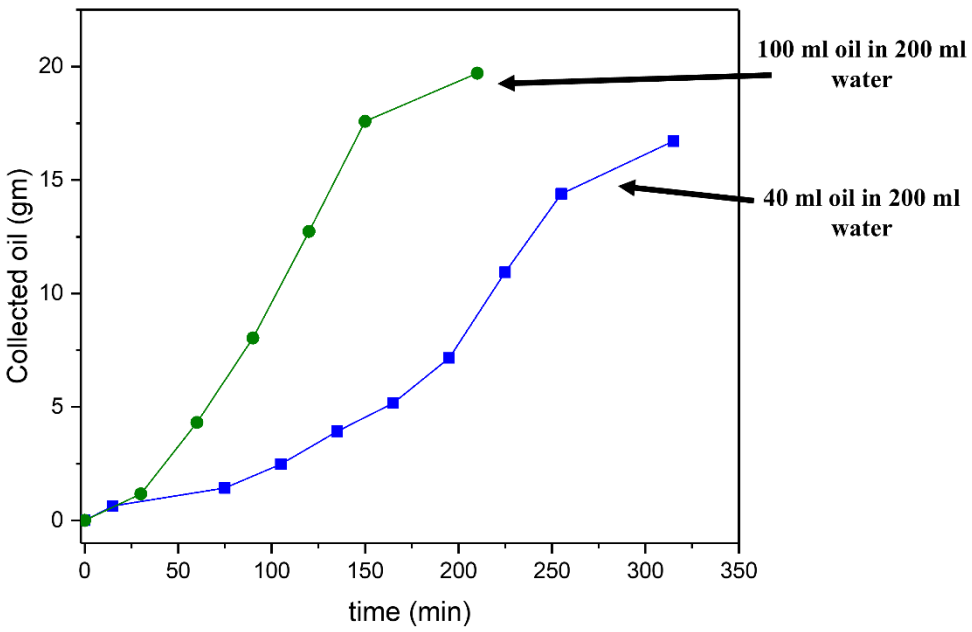


Figure 5. Setup images and plot between collected oil and time for 2 different concentrations.

2.5. Oil absorption capacity and oil extraction capacity

Figure 6 shows the oil absorption of coated sponge. Further oil absorption capacity is calculated using the below formula for both diesel and groundnut oil as shown in Table. The oil absorption capacity (κ) is calculated using the following equation. The calculated absorption capacity of the sponge for oil and diesel are 28.83 g/g and 23.15 g/g. Therefore, we can conclude that developed superhydrophobic has a very high absorption capacity and can be used for crude-oil water separation.

$$\kappa = \frac{M1 - M0}{M0}$$

where, M_0 = weight of sponge before oil absorption, M_1 = weight of sponge after it is saturated with oil

Table 2. Absorption capacity

Oil	**M_0 (gm)**	**M_1(gm)**	**κ (in gm/gm)**
Groundnut oil	0.438	13.067	28.83
Diesel	0.498	12.028	23.15

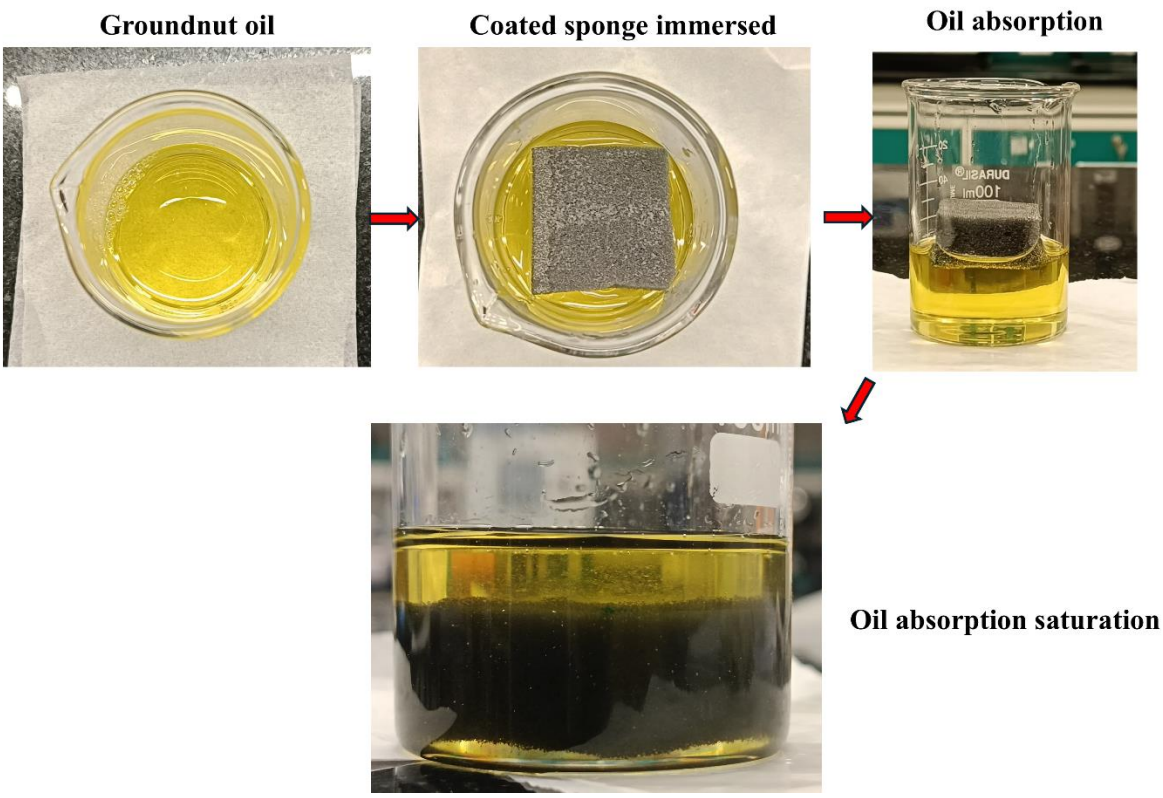


Figure 6. Setup to measure the oil absorption capacity of coated sponge.

2.6. Diesel removal from artificial salt water

Furthermore, the developed superhydrophobic sponge was tested for its ability to remove diesel from saltwater, simulating the separation of crude oil from seawater. Here, we first took the dyed saltwater in a petri dish, then Diesel is poured on it. Then, a coated sponge was placed over the diesel, after some time it completely absorbed the diesel. Further, we squeezed the oil from sponge and collected in petri dish as shown in the last image. Therefore, we can conclude that our developed coated sponge can be beneficial for the crude oil separation from the sea water.

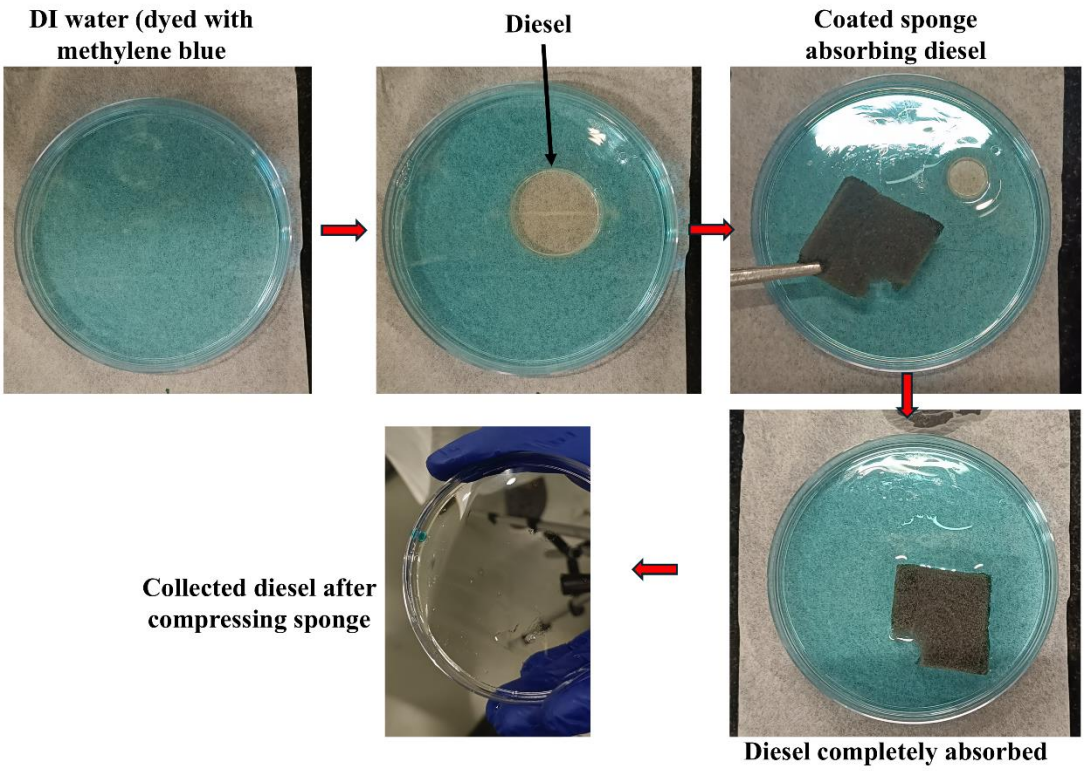


Figure 7. Setup for diesel removal from artificial salt water

2.7. Applying Lucas- Washburn equation for capillary driven flow for our system.

In this section, we applied the classical Lucas- Washburn (LW) [7] equation to understand the absorption of oil in the sponge. For horizontal tube, (inertial and gravity forces are neglected), the variation in imbibition length with time can be expressed by the following equation.

$$h(t) = \sqrt{\frac{r \cdot \sigma \cdot \cos\theta}{2 \cdot \mu}}$$

where, h = Height of the capillary rise, r = radius of the tube or pores, σ=surface tension of the oil, μ= viscosity of the wetting liquid, θ = contact angle of oil on the surface, t= time taken by the liquid to rise, ρ = density of the wetting liquid and g = acceleration due to gravity.

Further, the LW equation for the vertical capillary tube (considering gravity) can be expressed by the following equation

$$t(h) = -\frac{h}{B_h} - \frac{A_h}{B_h^2}\ln\left(1 - \frac{B_h h}{A_h}\right)$$

Where A_h and B_h is given by following expression

$$A_h = \frac{r\sigma \cos\theta}{4\mu}$$

$$B_h = \frac{\rho g r^2}{8\mu}$$

In our study, we used groundnut oil, therefore the values for different parameters in the equation as obtained form the literature are: σ = 33.33 x 10^{-3} N/m., μ = 53.14 x 10^{-3} Pa.s, θ = 87° (our study) r = 270 µm (our study), ρ =920 kg/m^3, g= 9.8 m/s^2

The values for the groundnut oil are plugged in the equations and the final expression for the sponge are expressed by the below equation. For horizontal tubes, which are relevant when the movement of oil can take place in the horizontal directions which is a possibility in our case. The obtained expression is as follows:

$$h(t) = 2.105 \times 10^{-3} \sqrt{t}$$

For vertical tubes, obtained expression is

$$t(h) = \frac{-h}{1.55\times10^{-4}} - 920\times\ln(1-0.07\times h)$$

Conclusion

From the study, we can conclude that developed superhydrophobic sponge has a very high absorption capacity of 28.83 g/g and 23.15 g/g for groundnut oil and diesel. Therefore, these sponges can be used to tackle the oil spillage problem in the sea.

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Individual contributions

Hanamanthagouda: Assistance in oil-water absorption experiments and other related experiments, literature survey on LW equation, original draft writing.

Ganivada Lalith: Assistance in oil-water absorption experiments and some other data collection.

Rakesh Choubey: Experiment planning, all experiments performed, literature survey for oil-water separation, report writing, original draft, review and editing

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